# An Effective Complex-Bimetric Negentropic-Sector Model for Dark Matter and Cosmic Acceleration

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We propose an effective complex-bimetric framework in which dark matter and cosmic acceleration arise as two distinct real gravitational projections of an orthogonal informational sector. The construction is based on a bimetric complex manifold $\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu}$, where $g_{\mu\nu}$ denotes the ordinary visible-sector spacetime metric and $f_{\mu\nu}$ denotes a negentropic metric associated with maintained informational order. The factor $i$ is not interpreted as a literal imaginary spacetime interval, but as a topological index encoding orthogonality between thermodynamic embodiment and informational geometry. The Standard Model is coupled minimally to $g_{\mu\nu}$, while a scalar-order sector and massive informational excitations propagate on $f_{\mu\nu}$. Localized excitations behave as pressureless dark matter, whereas the vacuum energy of the order parameter behaves as an effective negative-pressure component capable of sourcing cosmic acceleration. The effective graviton mass bridging the two metrics is dynamically generated via a Hassan–Rosen-type bimetric interaction, modulated by the negentropic scalar. We further introduce an emergy-inspired interpretation of the real sector and a complex value functional combining thermodynamic memory and informational order. The resulting model is presented as a phenomenological effective field theory, with consistency conditions from ghost freedom, stability, gravitational-wave propagation, and cosmological observables.

## I. INTRODUCTION

The standard cosmological model, ΛCDM, successfully describes a wide range of observations using two dominant dark components: cold dark matter and dark energy. Dark matter behaves as a pressureless clustering component, while dark energy is consistent with a nearly homogeneous component with equation of state $w \simeq -1$. Despite this phenomenological success, the microscopic origin of both components remains unknown.

A number of theoretical developments suggest that spacetime geometry, entropy, and information may be closely related. These include black-hole thermodynamics, holographic arguments, and emergent-gravity approaches. However, these frameworks do not by themselves provide a minimal phenomenological model capable of reproducing the observed dark sector while remaining testable.

In this work, we introduce an effective negentropic-sector model based on a bimetric structure, in which the physical metric $g_{\mu\nu}$ is supplemented by an additional metric $f_{\mu\nu}$. We formally group these into a bimetric complex manifold,

$$\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu}, \tag{1}$$

which serves as a compact representation of two coupled geometric sectors. Physical observables remain real and are associated with $g_{\mu\nu}$, while $f_{\mu\nu}$ encodes an effective orthogonal informational sector.

The negentropic sector contains two dominant degrees of freedom: a massive excitation $X$, which behaves as cold dark matter at the background level, and a slowly evolving scalar field $\chi$, whose potential energy can generate accelerated expansion. The model is therefore a minimal extension of ghost-free bimetric constructions, augmented by an interpretive framework in which the $f_{\mu\nu}$ manifold is associated with maintained informational order.

To make this interpretation operational, we frame the orthogonal informational sector as a macroscopic dissipative structure. Drawing on non-equilibrium thermodynamics, we posit that irreversible physical work in the visible sector stabilizes persistent informational states in the orthogonal sector. The maintenance of this order provides a geometric mechanism for the effective mass of the bimetric graviton.

The goal of this paper is twofold: first, to define a minimal effective bimetric model capable of reproducing the ΛCDM dark sector dynamically; and second, to formulate a generalized thermodynamic framework where the persistent order of the informational metric is sustained by the continuous dissipation of physical exergy.

## II. COMPLEX-BIMETRIC GEOMETRIC STRUCTURE

We introduce two Lorentzian metrics:

$$g_{\mu\nu} \qquad \text{and} \qquad f_{\mu\nu}. \tag{2}$$

The metric $g_{\mu\nu}$ is the physical metric seen by Standard Model matter. The metric $f_{\mu\nu}$ is the negentropic-sector metric.

The bimetric complex manifold is defined formally as

$$\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu}. \tag{3}$$

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The corresponding inverse and determinant are not assumed to define a directly measurable complex spacetime. Instead, the complex notation is used to encode the two-sector structure compactly. Physical observables are real-valued functionals obtained by projection,

$$\mathcal{O}_{\text{phys}} = \text{Re}\left[\mathcal{O}\left(\mathcal{G}_{\mu\nu}\right)\right]. \tag{4}$$

The ordinary spacetime interval is

$$ds_g^2 = g_{\mu\nu} dx^\mu dx^\nu, \tag{5}$$

while the informational-sector interval is

$$ds_f^2 = f_{\mu\nu} dx^\mu dx^\nu. \tag{6}$$

No Standard Model particle is assumed to follow geodesics of $f_{\mu\nu}$. The orthogonal informational sector couples to visible matter only gravitationally or through small portal interactions.

## III. ACTION

The full effective action is written as

$$S = S_g + S_f + S_{\text{mix}} + S_{\text{SM}} + S_{\mathcal{I}} + S_{\text{portal}}. \tag{7}$$

The visible gravitational action is

$$S_g = \frac{M_g^2}{2} \int d^4x \sqrt{-g} R[g]. \tag{8}$$

The informational gravitational action is

$$S_f = \frac{M_f^2}{2} \int d^4x \sqrt{-f} R[f]. \tag{9}$$

The Standard Model action is minimally coupled to $g_{\mu\nu}$,

$$S_{\text{SM}} = \int d^4x \sqrt{-g} \mathcal{L}_{\text{SM}}\left(\psi_{\text{SM}}, g_{\mu\nu}\right). \tag{10}$$

The orthogonal information-sector action is

$$S_{\mathcal{I}} = \int d^4x \sqrt{-f} \left[-\frac{1}{2} f^{\mu\nu} \nabla_\mu^{(f)} \chi \nabla_\nu^{(f)} \chi - U(\chi) + \mathcal{L}_X\right], \tag{11}$$

where $\chi$ is a slowly varying negentropic order parameter and $X$ denotes massive informational excitations.

A minimal example for the massive field is

$$\mathcal{L}_X = -\frac{1}{2} f^{\mu\nu} \nabla_\mu^{(f)} X \nabla_\nu^{(f)} X - \frac{1}{2} m_X^2 X^2 - \frac{\lambda_X}{4} X^4. \tag{12}$$

The bimetric mixing action is chosen to be of the Hassan–Rosen type to avoid the Boulware–Deser ghost at the effective level. Crucially, we promote the bimetric interaction scale to a dynamical function of the order parameter $\chi$, such that the coupling between the two metrics is mediated by the density of maintained informational order:

$$S_{\text{mix}} = m^2(\chi) M_{\text{eff}}^2 \int d^4x \sqrt{-g} \sum_{n=0}^{4} \beta_n e_n \left(\sqrt{g^{-1} f}\right), \tag{13}$$

where $e_n$ are the elementary symmetric polynomials of the matrix square root $\sqrt{g^{-1} f}$. In this framework, the effective mass of the massive spin-2 excitation is dynamically governed by the negentropic order parameter $\chi$. We parameterize this coupling via a conformal-like exponential factor, commonly employed in scalar-tensor theories:

$$m^2(\chi) = m_0^2 \exp\left(-\frac{\beta \chi}{M_{\text{Pl}}}\right), \tag{14}$$

where $m_0$ is the bare graviton mass and $\beta$ is a dimensionless coupling constant. Through this relation, the escalation of informational order dynamically modulates the interaction strength between the physical and negentropic metrics, physically manifesting the gravitational consequence of informational inertia.

The portal action is assumed to be small,

$$S_{\text{portal}} = \int d^4x \sqrt{-g} \left[-\frac{\lambda_{H\chi}}{2} H^\dagger H \chi^2 - \frac{\lambda_{HX}}{2} H^\dagger H X^2 + \cdots\right]. \tag{15}$$

The smallness of $\lambda_{H\chi}$ and $\lambda_{HX}$ explains why the orthogonal informational sector is not directly observed in laboratory physics.

## IV. FIELD EQUATIONS

Variation with respect to $g_{\mu\nu}$ yields

$$M_g^2 G_{\mu\nu}[g] = T_{\mu\nu}^{\text{SM}} + T_{\mu\nu}^{\text{portal}} + T_{\mu\nu}^{\text{mix},g}. \tag{16}$$

Variation with respect to $f_{\mu\nu}$ yields

$$M_f^2 G_{\mu\nu}[f] = T_{\mu\nu}^{\mathcal{I}} + T_{\mu\nu}^{\text{mix},f}. \tag{17}$$

The informational-sector stress-energy tensor is

$$T_{\mu\nu}^{\mathcal{I}} = \nabla_\mu^{(f)} \chi \nabla_\nu^{(f)} \chi - f_{\mu\nu} \left[\frac{1}{2} f^{\alpha\beta} \nabla_\alpha^{(f)} \chi \nabla_\beta^{(f)} \chi + U(\chi)\right] + T_{\mu\nu}^X. \tag{18}$$

Because the bimetric mixing scale $m^2(\chi)$ is dynamically coupled to the order parameter, the variation of $S_{\rm mix}$ with respect to $\chi$ yields a critical effective source term. The scalar order field obeys:

$$\Box_f \chi - \frac{dU}{d\chi} + \frac{1}{\sqrt{-f}} \frac{\delta S_{\rm mix}}{\delta \chi} = J_{\rm portal}, \tag{19}$$

where $J_{\rm portal}$ is a small source induced by portal couplings. The mixing derivative acts as a geometric restorative force, explicitly demonstrating that the cosmological interaction between the visible and negentropic metrics actively drives the dynamics of the informational vacuum, thereby stabilizing the cosmic acceleration.

The informational excitation $X$ obeys

$$\Box_f X - m_X^2 X - \lambda_X X^3 = J_X. \tag{20}$$

In the limit of negligible portal interactions, $J_{\rm portal} \approx 0$ and $J_X \approx 0$.

## V. COSMOLOGICAL BACKGROUND

We consider homogeneous and isotropic backgrounds,

$$ds_g^2 = -dt^2 + a^2(t) d\vec{x}^{\,2}, \tag{21}$$

$$ds_f^2 = -N_f^2(t) dt^2 + b^2(t) d\vec{x}^{\,2}. \tag{22}$$

The visible Hubble rate is

$$H_g = \frac{\dot{a}}{a}. \tag{23}$$

The informational-sector Hubble rate is

$$H_f = \frac{1}{N_f} \frac{\dot{b}}{b}. \tag{24}$$

The visible Friedmann equation takes the schematic form

$$3M_g^2 H_g^2 = \rho_b + \rho_r + \rho_{\rm DM}^{\rm eff} + \rho_{\rm DE}^{\rm eff} + \rho_{\rm mix}. \tag{25}$$

The effective dark matter density is associated with massive informational excitations,

$$\rho_{\rm DM}^{\rm eff} \equiv \rho_X^{\rm proj}. \tag{26}$$

For nonrelativistic excitations,

$$p_X \ll \rho_X, \tag{27}$$

and therefore

$$\rho_X \propto b^{-3}. \tag{28}$$

When projected into the visible sector, this behaves approximately as cold dark matter,

$$\rho_{\rm DM}^{\rm eff} \propto a^{-3}. \tag{29}$$

The effective dark energy density is associated with the negentropic order field,

$$\rho_{\rm DE}^{\rm eff} \equiv \rho_\chi^{\rm proj}. \tag{30}$$

For a slowly rolling or quasi-frozen order parameter,

$$\dot{\chi}^2 \ll U(\chi), \tag{31}$$

one obtains

$$\rho_\chi \approx U(\chi), \tag{32}$$

$$p_\chi \approx -U(\chi), \tag{33}$$

and hence

$$w_\chi = \frac{p_\chi}{\rho_\chi} \approx -1. \tag{34}$$

Thus the same orthogonal informational sector can generate both dark-matter-like and dark-energy-like behavior, provided its localized excitations and vacuum/order components are dynamically distinct.

## VI. DARK MATTER AS INFORMATIONAL INERTIA

In this model, dark matter is not interpreted as ordinary invisible matter, but as the visible gravitational projection of localized negentropic excitations.

The field $X$ carries energy density,

$$\rho_X = \frac{1}{2N_f^2} \dot{X}^2 + \frac{1}{2} m_X^2 X^2 + \frac{\lambda_X}{4} X^4. \tag{35}$$

When $X$ oscillates coherently in a quadratic potential, its averaged pressure vanishes,

$$\langle p_X \rangle \approx 0, \tag{36}$$

and therefore

$$\langle \rho_X \rangle \propto b^{-3}. \tag{37}$$

The visible sector feels this component through the bimetric interaction and gravitational projection. We define

$$\rho_{\rm DM}^{\rm eff} = \Gamma_X(a, b, \beta_n)\rho_X, \tag{38}$$

where $\Gamma_X$ is a projection factor depending on the relative evolution of the two metrics and on the bimetric coupling parameters.

Physically, $X$ may be interpreted as localized informational inertia. Because the $f_{\mu\nu}$ manifold operates in a regime of negative entropy, localized data structures resist deformation. This resistance to entropic decay manifests in the Hassan–Rosen mixing term as an effective inertial mass. Thus, $X$ constitutes an excitation that does not couple electromagnetically, yet contributes a restorative gravitational pressure—observed phenomenologically as Cold Dark Matter—to the visible universe.

## VII. COSMIC ACCELERATION FROM INFORMATIONAL ORDER

The scalar order parameter $\chi$ contributes

$$\rho_\chi = \frac{1}{2N_f^2}\dot{\chi}^2 + U(\chi), \tag{39}$$

$$p_\chi = \frac{1}{2N_f^2}\dot{\chi}^2 - U(\chi). \tag{40}$$

The equation-of-state parameter is

$$w_\chi = \frac{\frac{1}{2N_f^2}\dot{\chi}^2 - U(\chi)}{\frac{1}{2N_f^2}\dot{\chi}^2 + U(\chi)}. \tag{41}$$

If the field is trapped in a quasi-stable order phase,

$$\dot{\chi}^2 \ll N_f^2 U(\chi), \tag{42}$$

then

$$w_\chi \approx -1. \tag{43}$$

The visible effective dark-energy density is

$$\rho_{\rm DE}^{\rm eff} = \Gamma_\chi(a, b, \beta_n)\rho_\chi + \rho_{\rm mix}^\Lambda, \tag{44}$$

where $\rho_{\rm mix}^\Lambda$ denotes the part of the bimetric mixing potential that behaves as an effective cosmological constant.

In the interpretive language of this paper, cosmic acceleration is sourced by the pressure of maintained informational order.

## VIII. EMERGY, INFORMATION ORTHOGONALITY, AND COMPLEX VALUE

The complex notation used above admits a philosophical interpretation closely related to the elementary complex relation

$$z = a + ib. \tag{45}$$

The real component $a$ corresponds to embodied thermodynamic reality: matter, radiation, biological organization, geochemical structure, and the accumulated memory of solar work. The imaginary component $ib$ corresponds to orthogonal informational order.

This section does not introduce additional fundamental dynamics. Instead, it provides an interpretive layer connecting the effective field theory to a broader thermodynamic-information picture.

### A. Emergy and the Real Axis

We introduce a coarse-grained emergy density $\mathcal{E}_\odot$, representing history-dependent solar work stored in physical and biological structures. A phenomenological definition is

$$\mathcal{E}_\odot(x, t) = \int_{-\infty}^{t} K(t, t')\Phi_\odot(x, t')dt', \tag{46}$$

where $\Phi_\odot$ is absorbed solar-energy flux and $K(t, t')$ is a memory kernel encoding persistence, transformation efficiency, and ecological or geological storage.

The real-axis value density may then be represented schematically as

$$a(x, t) \sim \mathcal{E}_\odot(x, t). \tag{47}$$

This identification is not meant to replace stress-energy in general relativity. Rather, it provides a thermodynamic interpretation of the physical sector as a reservoir of embodied work.

At the effective level, one may write

$$T_{\mu\nu}^{\rm real} = T_{\mu\nu}^{\rm SM} + T_{\mu\nu}^{\rm bio} + T_{\mu\nu}^{\rm emergy}, \tag{48}$$

where $T_{\mu\nu}^{\rm bio}$ and $T_{\mu\nu}^{\rm emergy}$ are coarse-grained descriptions valid only at macroscopic ecological or planetary scales.

### B. Information Orthogonality and the Imaginary Axis

The metric $f_{\mu\nu}$ is interpreted as the geometric carrier of orthogonal informational order. The factor $i$ in

$$\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu} \tag{49}$$

denotes a $\pi/2$-like rotation in an abstract value plane. It does not imply that physical spacetime intervals are imaginary.

We define a coarse-grained information-order density,

$$\mathcal{I} = \frac{1}{V_f} \int_{\Sigma_f} d^3x \sqrt{\gamma_f} \left[ \frac{1}{2N_f^2} \dot{\chi}^2 + U(\chi) \right], \tag{50}$$

where $V_f$ is the informational-sector spatial volume,

$$V_f = \int_{\Sigma_f} d^3x \sqrt{\gamma_f}. \tag{51}$$

A phenomenological projection from real thermodynamic work into orthogonal information order may be written as

$$\mathcal{I} = \kappa_I \Pi_\perp \left[ \mathcal{E}_\odot \right], \tag{52}$$

where $\Pi_\perp$ is an abstract orthogonalization map and $\kappa_I$ is an emergy-information conversion coefficient.

This expression may be understood as a formal analogue of irreversible physical work being encoded into durable information. Cryptographic proof systems, including proof-of-work architectures, provide a terrestrial analogy: they transform dissipated physical work into an ordered, difficult-to-alter informational record. In the present cosmological model, however, such systems are not assumed to be the source of dark matter or dark energy. They serve only as an illustrative analogue of the more general principle of thermodynamic work projected into maintained information order.

### C. Negative Entropy as the Source of Hidden Curvature

The maintenance of macroscopic order requires the continuous export of entropy, establishing the orthogonal informational sector as an open dissipative structure. Following the Brillouin-Schrödinger paradigm, information acts as negative entropy (negentropy). The global thermodynamic constraint across the bimetric manifold requires:

$$dS_{\text{tot}} = dS_g + dS_f \geq 0, \tag{53}$$

where $S_g$ and $S_f$ are the entropies associated with the physical and negentropic metrics, respectively.

For the informational sector to progressively crystallize state updates, its local entropy evolution must be strictly negative:

$$\dot{S}_f < 0. \tag{54}$$

This violation of local entropy maximization is dynamically permitted only because it is coupled to massive, compensating thermal dissipation in the visible metric:

$$\dot{S}_g \gg \left| \dot{S}_f \right|. \tag{55}$$

This fundamental asymmetry has profound geometric consequences. In standard relativity, positive entropy production is associated with the dispersion of mass-energy. Conversely, the accumulation of negative entropy ($\dot{S}_f < 0$) within the informational metric implies an escalating local ordering.

We formally associate the scalar order field $\chi$ with the negentropic density of the orthogonal manifold:

$$\chi \propto \exp\left( -\frac{S_f}{k_B} \right). \tag{56}$$

Consequently, the negentropic sector does not violate the second law of thermodynamics; rather, it inverts its dynamical consequence. The negative entropy of the $f_{\mu\nu}$ metric acts as the fundamental source for the stress-energy tensor $T^{\mathcal{I}}_{\mu\nu}$, exerting a topological pressure that projects onto the visible universe as dark energy.

### D. Complex Value Functional

The philosophical norm associated with

$$z = a + ib \tag{57}$$

is

$$|z| = \sqrt{a^2 + b^2}. \tag{58}$$

In the present framework, the analogous real observable is a complex value functional combining thermodynamic embodiment and informational order:

$$\mathcal{V}_{\text{HB}} = \int_{\Sigma_t} d^3x \sqrt{\gamma_g} \left[ \mathcal{E}_\odot^2 + \alpha_I \mathcal{I}^2 \right]^{1/2}. \tag{59}$$

Here $\alpha_I$ converts information-order density into the same effective units as emergy density.

The functional $\mathcal{V}_{\text{HB}}$ is not a new fundamental interaction. It is a macroscopic diagnostic of joint physical-informational value. It formalizes the idea that a stable civilization, biosphere, or complex system cannot maximize informational abstraction while destroying its real thermodynamic base.

One may express a sustainability condition as

$$\frac{d\mathcal{V}_{\text{HB}}}{dt} > 0, \qquad \frac{d}{dt} \int_{\Sigma_t} d^3x \sqrt{\gamma_g} \mathcal{E}_\odot \geq 0. \tag{60}$$

The first condition states that total complex value grows. The second states that such growth must not be achieved by depleting the real-axis emergy base.

This provides a formal place for the philosophical concept of *Homo Biodiversitas*: an observer or civilization that increases informational order while preserving or regenerating the thermodynamic and ecological substrate from which that order is derived.

## IX. THERMODYNAMIC DISSIPATION AND IRREVERSIBLE STATE TRANSITIONS

The orthogonal informational sector introduced above acts as an effective geometric description of maintained macroscopic order. To ground this concept phenomenologically, it is instructive to examine systems where irreversible physical work is systematically converted into persistent, orthogonal informational structures. Distributed cryptographic consensus mechanisms provide a rigorous, albeit localized, thermodynamic analogue.

We stress that Proof-of-Work plays no dynamical role in the cosmological model. Its energy scale is negligible in comparison to cosmological densities, and no direct contribution to gravitational dynamics is implied. Its role is purely illustrative: it provides a controlled example of the mapping

$$\text{irreversible work} \longrightarrow \text{persistent information}. \tag{61}$$

Let $\mathcal{P}_{\rm diss}(t)$ denote the total exergy dissipation rate of a localized, fault-tolerant information network. The cumulative dissipated physical energy is

$$E_{\rm diss}(t) = \int_{t_0}^{t} \mathcal{P}_{\rm diss}(t')\, dt'. \tag{62}$$

The accumulated work securing the ledger can be represented by a monotonic quantity $W_{\rm cum}(t)$, proportional to the integrated difficulty-weighted computational effort. A dimensionless measure of persistent information order is then

$$\mathcal{I}_{\rm PoW}(t) = \log\left(1 + \frac{W_{\rm cum}(t)}{W_0}\right), \tag{63}$$

where $W_0$ is a reference scale.

The logarithmic form reflects the fact that security is governed by relative cost: each additional unit of accumulated work increases the difficulty of rewriting the ledger by a multiplicative factor.

The rate of order accumulation is

$$\dot{\mathcal{I}}_{\rm PoW} = \frac{\dot{W}_{\rm cum}}{W_0 + W_{\rm cum}}. \tag{64}$$

This construction highlights a key feature: the system generates an effective arrow of time. Past states become increasingly resistant to modification as cumulative work grows.

An effective thermodynamic cost of persistent memory may be defined as

$$\mathcal{C}_{\rm mem} = \frac{E_{\rm diss}}{\mathcal{N}_{\rm state}}, \tag{65}$$

where $\mathcal{N}_{\rm state}$ denotes the number of cryptographically sealed macroscopic state transitions within the topological register.

The analogy with the complex-value construction introduced earlier can be expressed through a localized functional:

$$\mathcal{V}_{\rm cons} = \left[\mathcal{E}_{\odot}^2 + \alpha_I \mathcal{I}_{\rm cons}^2\right]^{1/2}, \tag{66}$$

where $\mathcal{I}_{\rm cons}$ isolates the informational entropy reduction achieved through distributed consensus.

This expression mirrors the general structure

$$|z| = \sqrt{a^2 + b^2}, \tag{67}$$

with thermodynamic embodiment and informational order contributing as orthogonal components. Fundamentally, physical energy does not strictly encode information in its own native units. Instead, we invoke a macroscopic analogue to Landauer's Principle [24]. While Landauer established the minimum thermal dissipation required to erase a bit, our bimetric interaction operates as the inverse generative process: the continuous, irreversible dissipation of exergy in $g_{\mu\nu}$ is the required physical cost to crystallize and seal topological information within $f_{\mu\nu}$ [28]. Recent phenomenological frameworks, such as the D-Trinity protocol combining cryptographic hashing (SHA-256) with decentralized relays, provide a topological blueprint for how such rigid informational ledgers achieve macroscopic thermodynamic stability and phase transitions [29].

In this sense, Proof-of-Work consensus mechanisms provide a localized laboratory-scale example of the general principle underlying the orthogonal informational sector: persistent informational order requires a physical substrate, an energetic cost, and an irreversibility mechanism. This analogy is strictly operational and does not imply any direct cosmological role.

### A. Phenomenological Friction and the Bimetric Landauer Limit

In standard information theory, Landauer's principle defines the minimum thermal cost of erasing a bit [24]. We postulate an inverted, macroscopic analogue: the crystallization of topological order requires a continuous, irreversible expenditure of physical exergy.

Let $\delta W_{\rm diss}$ be the irreversible physical work dissipated in the visible metric $g_{\mu\nu}$, and $\delta\chi$ be the corresponding increment of negentropic order in $f_{\mu\nu}$. We hypothesize a fundamental threshold relation:

$$\delta W_{\rm diss} = \mathcal{T}^* \delta\chi, \tag{68}$$

where $\mathcal{T}^*$ acts as an effective topological tension (with dimensions of energy per unit order), representing the critical thermodynamic cost required to rigidly anchor an informational state against natural entropic decay.

In the language of field theory, the spontaneous evolution of the order parameter $\chi$ is governed by the Euler-Lagrange equation. To maintain the system away from thermal equilibrium (preventing the decay of $\chi$ down its potential $U$), the physical dissipation must act as an effective restorative force. We capture this by introducing a dissipative current $J_{\rm diss}$ into the scalar equation of motion:

$$\Box_f \chi - \frac{dU}{d\chi} + \frac{1}{\sqrt{-f}} \frac{\delta S_{\rm mix}}{\delta \chi} = J_{\rm diss}. \tag{69}$$

We model this current covariantly as being strictly sourced by the entropy production rate $\Phi_g$ within the physical metric. Taking $n^\lambda$ as the 4-velocity of the macroscopic informational flow, the friction current is:

$$J_{\rm diss} = \gamma(\Phi_g)\, n^\lambda \nabla_\lambda \chi, \tag{70}$$

where $\gamma(\Phi_g)$ is the bimetric friction coefficient, strictly monotonically increasing with the physical power dissipated in $g_{\mu\nu}$.

In the rest frame of the homogeneous background, $n^\lambda = (1,0,0,0)$, substituting the Hassan–Rosen interaction derivative yields the complete effective friction equation:

$$\Box_f \chi + \gamma(\Phi_g)\dot{\chi} - \frac{dU}{d\chi} + \frac{m_0^2}{M_{\rm Pl}} \beta \exp\left(-\frac{\beta\chi}{M_{\rm Pl}}\right) e_n\left(\sqrt{g^{-1}f}\right) = 0. \tag{71}$$

This equation dictates the phase transition of the negentropic sector. If the physical dissipation $\Phi_g$ is below a critical threshold, the friction term $\gamma(\Phi_g)\dot{\chi}$ is insufficient to overcome the natural potential gradient $dU/d\chi$, leaving the informational state fluid and non-consensual. However, when $\Phi_g$ exceeds this threshold, the thermodynamic friction dominates. The field $\chi$ is dynamically frozen into a high-negentropy metastable state, generating the rigid topological pressure required to source both the effective graviton mass and late-time cosmic acceleration.

## X. MINIMAL EFFECTIVE COSMOLOGICAL SECTOR

To make the phenomenological content explicit, we introduce a minimal effective negentropic-sector action. The aim is not to derive a fundamental theory of the information metric $f_{\mu\nu}$, but to provide a controlled parameterization of its two leading cosmological effects: localized pressureless excitations and a slowly varying order condensate.

We write

$$S_{\rm hid} = \int d^4x \sqrt{-f} \Big[ -\frac{1}{2} f^{\mu\nu} \partial_\mu X \partial_\nu X - \frac{1}{2} m_X^2 X^2 - \frac{1}{2} f^{\mu\nu} \partial_\mu \chi \partial_\nu \chi - U(\chi) \Big], \tag{72}$$

where $X$ denotes a massive informational excitation and $\chi$ denotes a slowly evolving information-order parameter.

For a homogeneous Friedmann background, the energy density and pressure of $X$ are

$$\rho_X = \frac{1}{2}\dot{X}^2 + \frac{1}{2} m_X^2 X^2, \tag{73}$$

$$p_X = \frac{1}{2}\dot{X}^2 - \frac{1}{2} m_X^2 X^2. \tag{74}$$

If $m_X \gg H$, the field oscillates rapidly compared with the Hubble time. Averaging over oscillations gives

$$\langle p_X \rangle \simeq 0, \qquad \langle \rho_X \rangle \propto a^{-3}. \tag{75}$$

Thus the massive excitation behaves as cold dark matter at the background level.

The order field $\chi$ has

$$\rho_\chi = \frac{1}{2}\dot{\chi}^2 + U(\chi), \tag{76}$$

$$p_\chi = \frac{1}{2}\dot{\chi}^2 - U(\chi). \tag{77}$$

Its equation-of-state parameter is therefore

$$w_\chi = \frac{\frac{1}{2}\dot{\chi}^2 - U(\chi)}{\frac{1}{2}\dot{\chi}^2 + U(\chi)}. \tag{78}$$

In the slow-order regime,

$$\dot{\chi}^2 \ll U(\chi), \tag{79}$$

one obtains

$$w_\chi \simeq -1, \tag{80}$$

so that the order sector behaves as an effective dark-energy component.

The visible-sector Friedmann equation may then be written as

$$3M_g^2 H^2 = \rho_{\rm SM} + \rho_X^{\rm eff} + \rho_\chi^{\rm eff} + \rho_{\rm int}, \tag{81}$$

where $\rho_{\rm int}$ encodes possible interaction or mixing terms between $g_{\mu\nu}$ and $f_{\mu\nu}$. At the level of background cosmology, the model therefore reproduces the standard decomposition

$$\rho_{\rm tot} = \rho_{\rm baryon} + \rho_{\rm radiation} + \rho_{\rm DM} + \rho_{\rm DE}, \tag{82}$$

with

$$\rho_{\rm DM} \equiv \rho_X^{\rm eff}, \qquad \rho_{\rm DE} \equiv \rho_\chi^{\rm eff}. \tag{83}$$

This construction is deliberately conservative. It does not require the informational sector to be directly identified with known digital or biological systems. Instead, the fields $X$ and $\chi$ are effective degrees of freedom whose phenomenology may be constrained by standard cosmological observations.

## XI. OBSERVATIONAL CONSTRAINTS AND FALSIFIABILITY

The model is intended as an effective parameterization and must therefore be constrained by the same observations that constrain ordinary dark matter, quintessence, and bimetric gravity. At minimum, a viable region of parameter space must satisfy the following conditions.

First, the effective dark-matter component must be sufficiently cold:

$$c_{s,X}^2 \ll 1 \tag{84}$$

during the epoch of structure formation.

Second, the order-field equation of state must remain close to that of a cosmological constant at late times:

$$w_\chi(z=0) = -1 + \epsilon, \qquad |\epsilon| \ll 1. \tag{85}$$

Third, the dynamic bimetric mixing dictates unique signatures in gravitational-wave (GW) astrophysics. The interaction between $g_{\mu\nu}$ and $f_{\mu\nu}$ induces graviton oscillations over cosmological distances. This implies a testable discrepancy between the GW luminosity distance ($d_L^{\rm GW}$) and the electromagnetic luminosity distance ($d_L^{\rm EM}$):

$$\frac{d_L^{\rm GW}(z)}{d_L^{\rm EM}(z)} \neq 1. \tag{86}$$

Current constraints from LIGO/Virgo multi-messenger events (e.g., GW170817) and future LISA observations place stringent bounds on the bare mass $m_0$ and the coupling $\beta$. Furthermore, the propagation speed of tensor perturbations must strictly satisfy $|(c_T^2/c^2) - 1| \lesssim 10^{-15}$.

Fourth, the effective Newtonian potential must reproduce standard large-scale structure (LSS) formation, governed by:

$$\nabla^2\Phi = 4\pi G_{\rm eff}(\chi, a)\left(\rho_{\rm baryon} + \rho_X^{\rm eff}\right). \tag{87}$$

Because $G_{\rm eff}$ is dynamically modulated by the informational order field $\chi$, the growth rate of structure, parameterized by $f\sigma_8$, will exhibit characteristic redshift-dependent deviations from the $\Lambda$CDM prediction.

Additionally, models with two independent metrics naturally source isocurvature perturbations. Data from the Planck satellite restricts the amplitude of Cold Dark Matter isocurvature modes to be tightly sub-dominant to adiabatic fluctuations, $\mathcal{P}_{\rm iso}/\mathcal{P}_{\rm ad} \ll 1$. The initialization of the $f_{\mu\nu}$ metric post-inflation must respect these cosmic microwave background (CMB) bounds.

These requirements make the model falsifiable. It is disfavored if the informational excitation suppresses structure formation, if the order field deviates strongly from $w \simeq -1$, or if the sector coupling induces observable deviations from general relativity.

| Parameter | Meaning | Phenomenological role |
|---|---|---|
| $m_X$ | excitation mass | controls cold-DM behaviour |
| $U(\chi)$ | order-field potential | controls dark-energy behaviour |
| $\alpha_I$ | information weight | weights orthogonal order |
| $W_0$ | reference work scale | normalizes PoW order |
| $\beta_{\rm mix}$ | sector mixing strength | controls deviations from GR |
| $M_f/M_g$ | Planck mass ratio | sets hidden gravity scale |

TABLE I. Minimal parameter set of the effective complex-bimetric model.

## XII. LINEAR PERTURBATIONS AND STABILITY CONDITIONS

A viable model must be stable at the perturbative level. We write scalar perturbations of the visible metric in Newtonian gauge,

$$ds_g^2 = -(1+2\Phi)dt^2 + a^2(t)(1-2\Psi)d\vec{x}^{\,2}. \tag{88}$$

For the $f_{\mu\nu}$ metric,

$$ds_f^2 = -N_f^2(1+2\Phi_f)dt^2 + b^2(t)(1-2\Psi_f)d\vec{x}^{\,2}. \tag{89}$$

The scalar perturbations are

$$\chi(t,\vec{x}) = \bar{\chi}(t) + \delta\chi(t,\vec{x}), \tag{90}$$

$$X(t,\vec{x}) = \bar{X}(t) + \delta X(t,\vec{x}). \tag{91}$$

The quadratic action for perturbations must take the schematic form

$$S^{(2)} = \frac{1}{2}\int dt\, d^3k\, \left[\dot{\vec{q}}^T K \dot{\vec{q}} - \vec{q}^T \Omega^2 \vec{q}\right], \tag{92}$$

where

$$\vec{q} = (\delta\chi, \delta X, \Phi, \Psi, \Phi_f, \Psi_f). \tag{93}$$

Ghost freedom requires the kinetic matrix to be positive definite,

$$K > 0. \tag{94}$$

Gradient stability requires positive sound speeds,

$$c_{s,i}^2 > 0. \tag{95}$$

Tachyonic instabilities are avoided when the relevant eigenvalues of the mass matrix satisfy

$$m^2_{\mathrm{eff},i} \gtrsim -H^2. \tag{96}$$

The tensor sector must also be stable. For tensor modes,

$$h^g_{ij}, \qquad h^f_{ij}, \tag{97}$$

the quadratic action takes the schematic form

$$S^{(2)}_T = \frac{1}{8}\int dt\, d^3k\, a^3 \Bigg[ M_g^2 \left( \dot{h}_g^2 - c_T^2 \frac{k^2}{a^2} h_g^2 \right) + M_f^2 \left( \dot{h}_f^2 - c_{T,f}^2 \frac{k^2}{b^2} h_f^2 \right) - m_T^2 (h_g - h_f)^2 \Bigg]. \tag{98}$$

Compatibility with gravitational-wave observations requires

$$c_T \approx 1 \tag{99}$$

in units where the speed of light is unity.

### A. Scalar Perturbations and Growth of Structure

To assess phenomenological viability, we consider linear scalar perturbations around a spatially flat Friedmann–Lemaître–Robertson–Walker background.

In Newtonian gauge, the visible metric is written as

$$ds_g^2 = -(1+2\Phi)dt^2 + a^2(t)(1-2\Psi)d\vec{x}^{\,2}. \tag{100}$$

We focus on sub-horizon scales, where $k \gg aH$, and assume that the negentropic sector couples weakly to visible matter. In this regime, the dominant effect of the orthogonal sector can be captured through an effective modification of the Poisson equation,

$$k^2\Phi = -4\pi G_{\mathrm{eff}}(k,a) a^2 \rho_m \delta_m. \tag{101}$$

The effective gravitational coupling may be parameterized as

$$G_{\mathrm{eff}}(k,a) = G\left[1 + \mu(k,a)\right], \tag{102}$$

where $\mu(k,a)$ encodes the contribution of the informational manifold.

For viable parameter ranges, we require

$$|\mu(k,a)| \ll 1 \tag{103}$$

at late times, ensuring consistency with large-scale structure and weak lensing observations.

The matter density contrast obeys

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\mathrm{eff}} \rho_m \delta_m = 0. \tag{104}$$

Thus, the orthogonal sector modifies structure growth through $G_{\mathrm{eff}}$. A small positive $\mu$ enhances clustering, while a negative value suppresses it.

The scalar order field $\chi$ contributes primarily at the background level through its effective equation of state,

$$w_\chi(z) = -1 + \delta w(z), \tag{105}$$

with

$$|\delta w(z)| \lesssim \mathcal{O}(10^{-2}) \tag{106}$$

required for consistency with current observational constraints.

Matching the observed dark-energy density implies

$$U(\chi_0) \sim (10^{-3}\,\mathrm{eV})^4, \tag{107}$$

which sets the characteristic scale of the order-field potential.

A full treatment would require solving the coupled perturbation system for $(\delta\chi, \delta X, \Phi, \Psi, \Phi_f, \Psi_f)$. However, the effective parameterization above captures the leading observable effects: a small modification of gravitational clustering and a nearly cosmological-constant background.

This provides a direct link between the $f_{\mu\nu}$ metric and observable quantities such as the growth rate $f\sigma_8$, weak lensing, and redshift-space distortions.

## XIII. INTERPRETATION

The model may be summarized by the correspondence

$$z = a + ib \quad \longleftrightarrow \quad \mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu}. \tag{108}$$

The real part $g_{\mu\nu}$ represents the geometry of thermodynamic embodiment: matter, radiation, chemistry, biology, and emergy-bearing structure. The imaginary part $if_{\mu\nu}$ represents an orthogonal informational geometry: topological order, memory, and non-electromagnetic structure.

Dark matter corresponds to localized informational inertia,

$$X \to \text{dark matter}. \tag{109}$$

Cosmic acceleration corresponds to global informational order pressure,

$$\chi \to \text{dark energy}. \tag{110}$$

The module-like quantity,

$$|z| = \sqrt{a^2 + b^2}, \tag{111}$$

is represented macroscopically by

$$\mathcal{V}_{\rm HB} = \int_{\Sigma_t} d^3x \sqrt{\gamma_g} \left[\mathcal{E}_\odot^2 + \alpha_I \mathcal{I}^2\right]^{1/2}. \tag{112}$$

This expression captures the central philosophical claim: reality is not exhausted by the material axis alone, nor by information alone, but by their normed synthesis.

## XIV. QUANTUM DECOHERENCE AS MICRO-BIMETRIC INTERACTION

While the macroscopic dynamics of the complex-bimetric framework explain dark sector phenomenology, the bimetric structure intrinsically accommodates the foundations of quantum mechanics. In standard quantum theory, the state of a system prior to measurement is described by a complex probability amplitude, $\psi(x,t) = \psi_R + i\psi_I$. Within our geometric correspondence, $\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu}$, this complex nature is no longer a purely statistical artifact but a reflection of the underlying bimetric topology.

We postulate that quantum superposition represents a regime where an excitation propagates predominantly within the informational metric $f_{\mu\nu}$, unconstrained by the macroscopic thermodynamic dissipation of the real axis. In this state, the particle explores the non-local topological graph of the informational sector, providing a geometric basis for quantum entanglement. Particles separated by vast spatial distances in $g_{\mu\nu}$ may remain topologically adjacent in $f_{\mu\nu}$.

The act of quantum measurement (decoherence) requires a macroscopic apparatus, inevitably involving irreversible thermodynamic dissipation in the visible metric. Thus, wavefunction collapse is interpreted as a micro-scale phase transition induced by the bimetric mixing potential $S_{\rm mix}$. The thermal friction of the measurement forces the delocalized informational state in $f_{\mu\nu}$ to precipitate onto the real axis $g_{\mu\nu}$. Decoherence is, therefore, the microscopic inverse of cryptographic consensus: thermal dissipation forcing a projection from the imaginary axis to the real axis. This geometric interpretation aligns with relational quantum mechanics and provides a unified origin for both quantum indeterminacy and cosmological dark matter.

## XV. LIMITATIONS

The present model is speculative and effective. It does not derive the metric $f_{\mu\nu}$ from a microscopic theory of quantum gravity. It also does not prove that information is literally geometric. Rather, it provides a phenomenological parameterization in which informational order is represented geometrically.

Several limitations must be emphasized.

First, the complex metric notation is formal. Physical observables remain real projections.

Second, the emergy and complex value functional are macroscopic interpretive constructs, rather than fundamental sources explicitly inserted into the Einstein field equations.

Third, the extension of this macroscopic bimetric framework to quantum decoherence remains phenomenological. The model does not yet provide a modified non-linear Schrödinger equation, nor a specific Lindblad master equation, to mathematically formalize how the bimetric mixing potential $S_{\rm mix}$ induces objective state reduction. Deriving the micro-scale energy dissipation of wave-function collapse strictly from the $f_{\mu\nu}$ metric requires a comprehensive quantum-gravitational treatment.

Fourth, localized cryptographic consensus architectures serve merely as rigorous thermodynamic analogies for the irreversible dissipation required to secure informational topology. They are not theorized to source cosmological dark matter.

Finally, a full non-linear perturbative analysis is required. The model must ultimately be confronted quantitatively with CMB isocurvature bounds, baryon acoustic oscillations, and gravitational-wave phase deviations.

## XVI. CONCLUSION

We have presented an effective complex-bimetric model in which the physical metric $g_{\mu\nu}$ is supplemented by an additional metric $f_{\mu\nu}$ describing an orthogonal negentropic sector. The resulting framework provides a minimal phenomenological description in which a massive excitation reproduces cold dark matter behavior, while a slowly evolving scalar field generates accelerated expansion.

The complex representation

$$\mathcal{G}_{\mu\nu} = g_{\mu\nu} + i f_{\mu\nu} \tag{113}$$

serves as a compact organizational tool rather than a literal extension of spacetime geometry. Physical observables remain associated with the real sector.

The model is constructed to be testable. Viability requires consistency with structure formation, cosmic expansion history, gravitational-wave constraints, and the absence of observable fifth forces. These conditions restrict the allowed parameter space of the negentropic sector.

An interpretive layer was introduced in which the $f_{\mu\nu}$ manifold is associated with maintained informational order. Proof-of-Work systems were used as a concrete analogue demonstrating how irreversible work can be converted into persistent informational structure. This analogy is strictly illustrative and does not imply any direct cosmological role.

Several open problems remain. A full perturbative analysis is required to establish stability and determine the detailed evolution of density perturbations. In addition, a deeper connection to fundamental theories of quantum gravity or statistical mechanics would be needed to derive the information sector from first principles.

Despite these limitations, the framework provides a controlled phenomenological setting in which dark matter, dark energy, and information-theoretic considerations can be discussed within a unified effective description.

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[1] A. Einstein, "Die Feldgleichungen der Gravitation," Sitzungsberichte der Preussischen Akademie der Wissenschaften zu Berlin, 844–847 (1915).
[2] E. Noether, "Invariante Variationsprobleme," Nachr. Ges. Wiss. Göttingen, Math.-Phys. Kl. 1918, 235–257 (1918).
[3] A. Friedmann, "Über die Krümmung des Raumes," Z. Phys. **10**, 377–386 (1922). DOI:10.1007/BF01332580
[4] G. Lemaître, "Un univers homogène de masse constante et de rayon croissant rendant compte de la vitesse radiale des nébuleuses extragalactiques," Ann. Soc. Sci. Bruxelles A **47**, 49–59 (1927). DOI:10.1093/mnras/91.5.483
[5] F. Zwicky, "Die Rotverschiebung von extragalaktischen Nebeln," Helv. Phys. Acta **6**, 110–127 (1933). DOI:10.1007/s10714-008-0707-4
[6] V. C. Rubin and W. K. Ford, "Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions," Astrophys. J. **159**, 379–403 (1970). DOI:10.1086/150317
[7] A. G. Riess et al., "Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant," Astron. J. **116**, 1009–1038 (1998). DOI:10.1086/300499
[8] S. Perlmutter et al., "Measurements of Omega and Lambda from 42 High-Redshift Supernovae," Astrophys. J. **517**, 565–586 (1999). DOI:10.1086/307221
[9] N. Aghanim et al., "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. **641**, A6 (2020). DOI:10.1051/0004-6361/201833910
[10] C. de Rham, G. Gabadadze, and A. J. Tolley, "Resummation of Massive Gravity," Phys. Rev. Lett. **106**, 231101 (2011). DOI:10.1103/PhysRevLett.106.231101
[11] S. F. Hassan and R. A. Rosen, "Bimetric Gravity from Ghost-free Massive Gravity," JHEP **02**, 126 (2012). DOI:10.1007/JHEP02(2012)126
[12] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, "Modified Gravity and Cosmology," Phys. Rept. **513**, 1–189 (2012). DOI:10.1016/j.physrep.2012.01.001
[13] G. Bertone, D. Hooper, and J. Silk, "Particle dark matter: Evidence, candidates and constraints," Phys. Rept. **405**, 279–390 (2005). DOI:10.1016/j.physrep.2004.08.031
[14] J. L. Feng, "Dark Matter Candidates from Particle Physics and Methods of Detection," Ann. Rev. Astron. Astrophys. **48**, 495–545 (2010). DOI:10.1146/annurev-astro-082708-101659
[15] E. J. Copeland, M. Sami, and S. Tsujikawa, "Dynamics of dark energy," Int. J. Mod. Phys. D **15**, 1753–1936 (2006). DOI:10.1142/S021827180600942X
[16] A. Joyce, B. Jain, J. Khoury, and M. Trodden, "Beyond the Cosmological Standard Model," Phys. Rept. **568**, 1–98 (2015). DOI:10.1016/j.physrep.2014.12.002
[17] J. D. Bekenstein, "Black holes and entropy," Phys. Rev. D **7**, 2333–2346 (1973). DOI:10.1103/PhysRevD.7.2333
[18] S. W. Hawking, "Particle Creation by Black Holes," Commun. Math. Phys. **43**, 199–220 (1975). DOI:10.1007/BF01608497
[19] J. A. Wheeler, "Information, physics, quantum: The search for links," in *Complexity, Entropy, and the Physics of Information*, edited by W. H. Zurek, Addison-Wesley (1990).
[20] W. H. Zurek, "Thermodynamic cost of computation, algorithmic complexity and the information metric," Nature **341**, 119–124 (1989). DOI:10.1038/341119a0
[21] N. Wiener, *Cybernetics: Or Control and Communication in the Animal and the Machine*, MIT Press (1948).
[22] E. Schrödinger, *What Is Life? The Physical Aspect of the Living Cell*, Cambridge University Press (1944).
[23] H. T. Odum, *Environmental Accounting: Emergy and Environmental Decision Making*, Wiley (1996).
[24] R. Landauer, "Irreversibility and Heat Generation in the Computing Process," IBM J. Res. Dev. **5**, 183–191 (1961). DOI:10.1147/rd.53.0183
[25] C. H. Bennett, "The thermodynamics of computation—a review," Int. J. Theor. Phys. **21**, 905–940 (1982). DOI:10.1007/BF02084158
[26] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," White paper (2008).
[27] E. P. Verlinde, "On the Origin of Gravity and the Laws of Newton," JHEP **04**, 029 (2011). DOI:10.1007/JHEP04(2011)029
[28] M. M. Vopson, "The second law of information dynamics," AIP Advances **12**, 075310 (2022). DOI:10.1063/5.0100358
[29] P. Ranaora and J. Yii, "Thermodynamic Stability and Phase Transitions in the Nakamoto Consensus," IPI Lett. (2026). DOI:10.59973/ipil.325
[30] R. Penrose, "On gravity's role in quantum state reduction," Gen. Relativ. Gravit. **28**, 581–600 (1996). DOI:10.1007/BF02105068
[31] C. Rovelli, "Relational quantum mechanics," Int. J. Theor. Phys. **35**, 1637–1678 (1996). DOI:10.1007/BF02302261